MIL-G-10272A (MU)
6 March 1964
SUPERSEDING
MIL-G-10272 (Ord)
28 April 1950

MILITARY SPECIFICATION

GROMMETS; METAL-FIBER (FOR PROJECTILE)

1. SCOPE

1.1 This specification covers one type of metal-fiber grommet for protecting rotating bands of projectiles.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-P-10025 - Packaging and Packing for Domestic Shipment of Inert Ammunition Components
MIL-I-45208 - Inspection Requirements; General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105D - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions

FSC 8140

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MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ORDNANCE CORPS

10520044 - Grommet Assembly for 155MM, 8 In. and 240MM
10520078 - Grommet Assembly

PUBLICATIONS

ORDNANCE CORPS

IEL 9200228 - Index of Inspection Equipment List

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Material and parts shall be in accordance with applicable specifications and drawings.

3.2 Grommets.-The grommet shall comply with all requirements specified on Drawings(dwgs) 10520044 and 10520078 and with all requirements specified in applicable specifications.

3.3 Impact

3.3.1 Flat blow.-The grommet shall protect the rotating band from damage when subjected to the impact of a 15 pound striker dropped from a height of 5 feet, when tested as specified in 4.4.1.

3.3.2 Edge blow.-The grommet shall protect the rotating band from damage when subjected to the impact of a 15 pound striker dropped from a height of 3 feet, when tested as specified in 4.4.1.

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3.4 Water absorption of the liner.-The liner shall absorb not more than 10 percent of water by weight, when tested as specified in 4.4.2.

3.5 Workmanship.-The grommet shall be free from cracks, chips, dirt, grease and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

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d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analysis on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection

4.2.1 Prior to the start of regular production, the contractor shall submit an initial production sample to a Government approved facility as designated by the Contracting Officer (see 6.4) for evaluation in accordance with the provisions of 4.3.2 and 4.3.3. The initial production sample shall consist of 25 grommets and 10 complete sets of parts which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and requirements which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis for materials. An initial production sample, or portion thereof,

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as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Components will be subjected by the Government to any or all of the examinations specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any component fails to comply with any or all of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component in the sample to comply with any or all of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. Until an initial production sample is accepted, the contractor is, in no way authorized by the Government to initiate regular production unless otherwise directed by the Contracting Officer.

4.3 Inspection provisions

4.3.1 Lot formation.-A lot shall consist of grommets produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting performance or fit, as determined by the Government, shall not necessitate changing the lot interfix number. Product shall be submitted for inspection in accordance with Standard MIL-STD-105D (or Standard MIL-STD-1235, when applicable).

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4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105D. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Housing, liner (see dwg. 10520045 covering a detail of dwg. 10520044 or dwg. 10520083 covering a detail of dwg. 10520078.

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	None defined		
Minor:	AQL 4.00 percent		
201.	Large diameter maximum, (max.)	Gage	01001
202.	Small diameter	Gage	01002
203.	Intermediate diameter	Gage	01003
204.	Width	Gage	01004
205.	Thickness	Gage	01005
206.	Profile of tie slots	Gage	01006
207.	Position of tie slots	Gage	01007
208.	Protective coating missing or with a total damaged area in excess of 1/4 square inch	Visual	01008
209.	Evidence of poor workmanship (see 3.5)	Visual	01009

4.3.2.2 Liner (see dwg. 10520046 covering a detail of dwg. 10520044 or dwg. 10520082 covering a detail of dwg. 10520078).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major: None defined

Minor: AQL 2.50 percent

201.	Large diameter	Gage	02001
202.	Small diameter	Gage	02002
203.	Thickness	Gage	02003
204.	Length of tie slot	Gage	02004
205.	Depth of tie slot	Gage	02005
206.	Liner arc missing	Visual	02006
207.	Evidence of poor workmanship (see 3.5)	Visual	02007

4.3.2.3 Spacer, (see dwg. 10520080 or 10520081 covering a detail of dwg. 10520078) (applicable to 175MM only).

Categories	Defects	Method of Inspection	Code No.

Critical: None defined

Major: None defined

Minor: AQL 1.50 percent

201.	Width	Gage	03001
202.	Arc missing	Visual	03002
203.	Evidence of poor workmanship (see 3.5)	Visual	03003

4.3.2.4 Liner assembly (see dwg. 10520079 covering a detail of dwg. 10520078) (applicable to 175MM only).

Categories	Defects	Method of Inspection	Code No.

Critical: None defined

Major: None defined

Minor: AQL 1.50 percent

201.	Spacer missing or improperly assembled	Visual-Gage	04001

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202.	Staple missing or improperly assembled	Visual	04002
203.	Crown or crimped end of staple above flush	Visual	04003
204.	Evidence of poor workmanship (see 3.5)	Visual	04004

4.3.2.5 Tie (see dwg. 10520062 covering a detail of 10520044 or 10520078).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.00 percent		
201.	Length	Gage	05001
202.	Evidence of poor workmanship (see 3.5)	Visual	05002

4.3.2.6 Grommet assembly (see dwg. 10520044 or 10520078).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 2.50 percent		
201.	Staple missing or above flush	Visual	06001
202.	Liner incorrectly assembled or insecure	Visual	06002
203.	Assembly damaged to the extent that its functional effectiveness has been destroyed	Visual	06003
204.	Protective coating with total damaged area in excess of 1/4 square inch	Visual	06004
205.	Evidence of poor workmanship (see 3.5)	Visual	06005

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4.3.3 Testing

4.3.3.1 Impact

4.3.3.1.1 Flat blow.-A random sample of 12 grommets shall be selected from the lot. If any shell rotating band is marred or material is left in the groove of the rotating band of 5 or more of the 48 blows, (a slight desposit or film of material from the liner left on the rotating band will not be considered objectionable) the lot shall be rejected. This test shall be performed as specified in 4.4.1 using equipment specified in 4.3.4.

4.3.3.1.2 Edge blow.-A random sample of 4 grommets shall be selected from the lot for this test. If any shell rotating band is marred or the steel housing of the grommet is punctured, or material is left in the grove of the rotating band of 3 or more of the 16 blows (a slight deposit of material from the liner left on the rotating band or breaking of the liner under the striker will not be considered objectionable) the lot shall be rejected. This test shall be performed as specified in 4.4.1 using equipment specified in 4.3.4.

4.3.3.2 Water absorption of the liner.-A random sample of 4 grommets shall be selected from the lot for this test. If any sample fail to comply with the requirement the lot shall be rejected. This test shall be performed as specified in 4.4.2.

4.3.4 Inspection Equipment.-Index of Inspection Equipment Lists Number IEL 9200228 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4.1 Impact

4.4.1.1 Flat blow.-The 12 grommets selected for this test shall be divided into 3 groups of 4 each. One group shall be conditioned at 75 degrees Fahrenheit (F.) plus or minus 5 degrees for 24 hours minimum before testing. The second groups shall be conditioned at 175 degrees (F.) plus or minus 5 degrees for 24 hours minimum before testing. The third group shall be conditioned at minus 40 degrees (F.) plus or minus 5 degrees for 24 hours minimum before testing. At the end of each conditioning period the grommets shall be assembled to a shell of the class on which the grommet will be used or on a simulated shell having a body and rotating band of materials and dimensions prescribed for the service shell. The grommet and shell assembly shall be placed in the fixture and a 15 pound striker shall be dropped from a height of 5 feet (see Applicable inspection equipment drawing referenced on IEL) to contact the grommet. After each blow the grommet and shell assembly shall be rotated on the longitudinal axis through approximately 90 degrees and another blow applied until a total of 4 blows have been applied. No blows shall contact a wire tie. The above procedure shall be repeated on each assembly until the 12 grommet and shell assemblies have been tested.

4.4.1.2 Edge blow.-The 4 grommets selected for this test shall be conditioned at 75 degrees (F.) plus or minus 5 degrees for 24 hours minimum before testing. At the end of this period the grommets shall be assembled to a shell of the class on which the grommet will be used or on a simulated shell having a body and rotating band of materials and dimensions prescribed for the service shell. The grommet and shell assembly shall be placed

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in the fixture and a 15 pound striker shall be dropped from a height of 3 feet (see applicable equipment drawing referenced on IEL) to contact the grommet. After each blow, the grommet and shell assembly shall be rotated on the longitudinal axis through approximately 90 degrees and another blow applied until a total of 4 blows have been applied. No blows shall contact a wire tie. The above procedure shall be repeated on each assembly until the 4 grommet and shell assemblies have been tested.

4.4.2 Water absorption of the liner.-The fiber liners shall be removed from the 4 grommets selected for this test. The fiber liners shall be placed in a drying oven maintained at a temperature of 100 degrees (F.) plus or minus 5 degrees for 48 hours. The liner shall then be placed in a conditioning chamber maintained at a temperature of 73 degrees (F.) plus or minus 2 degrees and a relative humidity of 50 percent plus or minus 2 percent. After 72 hours the liner shall be removed, each one immediately weighed, and then immersed under a head of approximately 2 inches of fresh tap water maintained at a temperature of 78 degrees (F.) plus or minus 2 degrees. After 24 hours immersion the liners shall be removed, the surface water quickly wiped off with a dry cloth, and each one immediately weighed. The amount of water absorbed by each liner shall be determined by subtracting the weight of the liner after removal from the conditioning chamber from its weight after immersion in water. The difference in weight shall be multiplied by 100 and divided by the weight of the liner after removal from the conditioning chamber. The result obtained will be the percentage of water absorption of the liner.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking

5.1.1 Level C.-The parts shall be packed and marked in accordance with Specification MIL-P-10025.

5.2 Data cards.-Data cards shall be prepared for each lot in accordance with information specified in Specification MIL-A-2550.

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6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Projectile for which the grommet is used.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

6.3.1 Inspection Equipment Lists.-Inspection Equipment Lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection Equipment List codes.-The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the Inspection Equipment List by the following codes:

CDOF - Contractor's design responsibility on Ordnance format; ORDM-608-12.
CDCF - Contractor's design responsibility on contractor format.
OD - Ordnance design.
ODMU - Ordnance design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control Drawing.

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6.3.3 Ordnance designs.-Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and non-mandatory. When an Inspection Equipment List references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Specification control drawings.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be

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supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608-12, and, in addition, contain the detail and information specified above.

6.3.7 Submission of contractor designs.-Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Submission of initial production samples.-Instructions as to the location for evaluation of initial production shall be obtained from the Contracting Officer. Upon receipt of such request, the Contracting Officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-NI, Dover, New Jersey.

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